Viscar Industrial Capacity Limited
Enhancing Performance & Productivity

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Programmable Logic Controllers (PLC) Level 1

The focus of this module is to equip the participants with PLC programming skills; so that they understand the tools of the software. It also enables the participants to develop knowledge of the hardware through wiring of the individual PLC modules, the module also looks at troubleshooting PLCs.

This workshop casts light to industrial motion control systems in regard to their design architecture, hardware components, parameterization and programming. This training course focuses on the concept of hardware and software together with practical hands-on introduction to uses, hardware configuration, uploading / downloading program and fault finding.

All topics are illustrated through practical exercises on training simulator using the Siemens PLC S7 300 series/Schneider/Telemecanique/ABB/Omron/Allen Bradley/B&R/ Mitsubishi.

The course will be structured around:

i. Hardware configuration; Components and wiring
ii. Software Configuration; Programming tool, communication/networking
iii. Creating programs, downloading and executing
iv. Practical sessions – Creating PLC Projects; Simulation of actual industrial problems
v. Fault diagnosis, troubleshooting and maintenance

Target Audience: Electrical, Mechatronics and Mechanical Engineers, Technicians

Duration: 1 Week

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Programmable Logic Controllers (PLC) Level 2

This course provides deep understanding in integrating PLC with others systems i.e. VFDs, Motors, Process Control. It entails application of the programming tools learnt in Level 1 to create programs that control other systems. It will cover PLC wiring, programming, diagnosis and troubleshooting systems that are controlled by PLCs.

All topics are addressed through practical illustrated exercises on training rigs using Siemens PLC S7-1500/ Schneider PLC and programing using Total Integrated Automation (TIA) /SOMachine Software

The workshop will focus on participants;

- Understanding how PLCs interface with other systems
- Understanding how PLCs control systems such as; VFDs, HMIs, High Speed Counters, Encoders
- Understand Analogue Systems Processing
- Understand Structured Programming
- Solving simple to complex problems with S7 -1200 controllers/Schneider PLC Series in order to minimize unplanned plant breakdowns and maximize plant utilization.
- Skills to avoid common problems related to the technology and appreciate the need for preventive maintenance for trouble free operations of our equipment
- Finding errors in complex set-ups quickly and efficiently.
- Understand PLC interface with SCADA Systems

Pre-requisite: Participants must demonstrate competency on Level 1

Target Audience: Electrical, Mechatronics and Mechanical Engineers, Technicians

Duration: 1 Week